

Integrating diverse evidence using Evidential Pluralism

Introductory Guide

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1. Evidential Pluralism: What it is and when to use it

Evidential Pluralism (EP) is a framework for understanding how to integrate different kinds of evidence when evaluating whether an intervention works. EP yields evaluations informed by *all* relevant evidence—quantitative and qualitative—not just a small subset of quantitative studies.

EP does this by assessing mechanistic evidence alongside the associative evidence provided by experimental and observational studies. Evidence that key features of the proposed mechanism of action of the intervention are operating as intended can raise confidence in effectiveness. On the other hand, if these features are found to be absent, or if key features of counteracting mechanisms are found to be present, this can undermine confidence in effectiveness. Either way, evidence of mechanisms can be highly informative. EP can help us to evaluate this evidence when assessing whether an intervention works.

EP can be thought of as a framework for integrating standard methods for systematically reviewing experimental and observational studies with theory-based methods, which focus on mechanisms. However, while theory-based evaluation designs such as realist evaluation usually seek to ask *how* an intervention works, EP focusses on *whether* the intervention works. Thus, EP provides a methodology for impact evaluation (assessing whether an intervention is an effective means of achieving certain goals), rather than process or implementation evaluation (assessing whether the intervention has been implemented as intended).

EP is particularly appropriate:

- where experimental and observational studies on their own fail to establish or rule out causation;
- where mechanistic considerations might inform the assessment of experimental and observational studies (e.g., by identifying potential confounding variables that have not been adequately controlled for);
- to ensure that qualitative evidence informed by stakeholders is not marginalised.¹

The application of EP to evidence-based medicine is called ‘EBM+’ and the application of EP to evidence-based policy is called ‘EBP+’. The ‘+’ denotes the addition of explicit procedures for evaluating mechanistic evidence, to be used together with the standard methods of orthodox evidence-based evaluation. This guide focusses on the area of policy evaluation, i.e., EBP+. §§2-3 introduce the overall approach. §4 uses an example to illustrate the key steps of an EP evaluation. §5 provides an example in which the cause being assessed is not an intervention. §§6-9 provide examples of the use of EP to identify gaps in the evidence base. §10 points to further resources and §11 provides a glossary of terminology.

2. Introducing Evidential Pluralism

Correlation is not causation. Why? Because a correlation between intervention variable *A* and outcome variable *B* (controlling for potential confounders) could be attributable to any one of a large number of possible explanations—of which causation is just one. For example, a correlation might be due to reverse causation (i.e., *B* causing *A*), uncontrolled confounding variables, or various kinds of bias introduced by the study design. If the correlation really is attributable to *A* being a cause of *B* then there must be some mechanism of action by which *A* produces *B*. So, in order to

¹ See Trofimov & Williamson (2026).

establish causation one needs to establish not only correlation but also the existence of an underlying mechanism that can account for the extent of the observed correlation. This observation motivates Evidential Pluralism, which is depicted in Fig. 1.

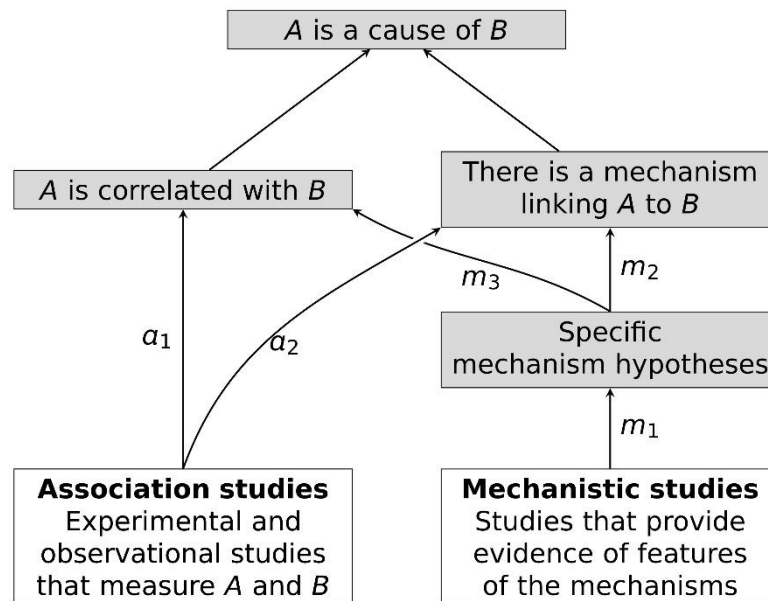


Figure 1. Evidential relationships for causal enquiry, according to Evidential Pluralism.

A purely quantitative approach to evaluation seeks to establish causation just by means of experimental and observational studies that repeatedly measure A and B together with potential confounders. Here, randomised controlled trials (RCTs) are particularly informative because they not only provide an estimate of the extent of the correlation (confirming correlation via channel a_1 in Fig. 1), but they also indirectly confirm the existence of an underlying mechanism (channel a_2) because they lower the probability that the correlation is attributable to uncontrolled confounders.

But the presence of an appropriate mechanism can be tested more directly by hypothesising specific features of the mechanism (m_2) and then exploring whether or not these features are actually present (m_1). Such features include key mediating variables, or entities and activities involved in the mechanism, or features of the way in which these entities and activities are organised. A study that looks for such features is classified as a ‘mechanistic study’.

A mechanistic study might use quantitative methods. For example, to test the hypothesis that some variable C mediates a mechanism between A and B, one might carry out an RCT that looks for a correlation between A and C and another quantitative study that looks for a correlation between C and B. A mechanistic study might, alternatively, use qualitative methods. For example, qualitative responses from interviews might be used to isolate features of the ways in which A and B are connected in the lives of individual stakeholders; qualitative comparative analysis might be used to test which features of a mechanism are key to its execution. Alternatively, a mechanistic study might use mixed methods: e.g., triangulation from quantitative and qualitative methods to establish the presence of some key feature of interest.

3. How to integrate diverse evidence

The EBP+ approach to integrating these streams of evidence is summarised in Table 1 below. See section 4 for an in-depth example that goes through the steps of Table 1.

Table 1. The EBP+ approach to evaluating the effectiveness of an intervention. This table is based on Shan & Williamson (2023, Table 3.3) and Trofimov & Williamson (2025, Table 2).

Task	Main questions
1. Specify the causal claim.	What is the cause, outcome, population and time-frame?
2. Identify specific mechanism hypotheses.	What are the key features of the mechanism by which the cause is supposed to operate? What are the key features of any enabling, enhancing and counteracting mechanisms?
3. Specify the correlation claim.	Which variables do the specific mechanism hypotheses identify as potential confounders? Are there any other potential confounders?
4. Identify and assess association studies.	What are the search terms? What are the inclusion criteria for the studies? How informative is each included study?
5. Assess the need for a mechanistic evaluation.	How well confirmed is the causal claim, given the association studies? Could evidence of specific mechanism hypotheses change this preliminary determination? (If so, proceed.)
6. Identify and assess mechanistic studies.	Which features of specific mechanism hypotheses have previously been established or ruled out? Which review questions should be used to find studies relevant to remaining features? How informative is each identified study?
7. Assess specific mechanism hypotheses.	How well confirmed is each specific mechanism hypothesis, given the mechanistic studies?
8. Assess the correlation claim.	How well confirmed is the correlation claim, given the specific mechanism hypotheses and the association studies?
9. Assess the general mechanistic claim.	How likely is it that there is some mechanism complex by which the cause produces the outcome, given the specific mechanism hypotheses and association studies? Can such a mechanism complex account for the magnitude and variation of the observed correlation?
10. Infer the status of the causal claim.	The causal claim is as well confirmed as the weakest of the correlation and general mechanistic claims.

EP can thus provide guidance on where and how to combine quantitative and qualitative evidence:

- Quantitative and qualitative evidence will need to be scrutinised in order to assess specific mechanism hypotheses (channel m_1 of Fig. 1; step 7 of Table 1).
- A second point of integration arises when ascertaining whether there is an appropriate mechanism linking A and B. Here, one needs to consider how plausible it is that some suitable mechanism exists, given quantitative evidence from experimental and observational studies (channel a_2) and the quantitative and qualitative evidence from mechanistic studies (m_2 ; step 9).
- A third point of integration arises when assessing correlation. Although quantitative studies are likely to be most informative here (a_1), mechanism hypotheses can suggest potential confounders that may not have been adequately controlled for, decreasing confidence that A and B are correlated conditional on potential confounders (m_3). Alternatively, if the mechanism is well established and sufficiently simple, the features of the mechanism may increase confidence in a correlation (m_3 ; step 8).

4. Example: Covid-19 face mask mandates

A narrow focus on experimental studies, especially RCTs, resulted in controversy and uncertainty concerning the effectiveness of public health interventions to reduce the spread of Covid-19, including legal requirements to wear a face mask in public. A Cochrane Systematic Review, which included only RCTs, found face masks make little or no difference to the spread of Covid-19.² Various limitations of the review and of the studies included have, however, been highlighted, with some authors arguing that a broader range of evidence is crucial to understanding the effectiveness of face masks.³ This case study therefore provides a good example of the need for and benefits of an EP evaluation. We sketch a proof-of-concept evaluation here, following the steps set out in Table 1. See Trofimov & Williamson (2025) for further details about this evaluation.

1. Specify the causal claim:

The causal claim is that a legal requirement to wear a face mask in public reduces the prevalence of symptomatic Covid-19 infections and thereby reduces the number of hospitalizations and deaths.

The population of interest is the global population. An evaluation of a specific law on reducing the prevalence of symptomatic Covid-19 infections will focus on the population of the jurisdiction within which the specific law applies. However, here we are concerned with the effectiveness of legal requirements to wear face masks at a more general level.

The legal requirements of interest are those that require a cloth face mask to be worn in public, especially when indoors or when outside and maintaining social distancing is difficult. Exemptions include relevant health conditions that make wearing a face mask problematic or difficult.

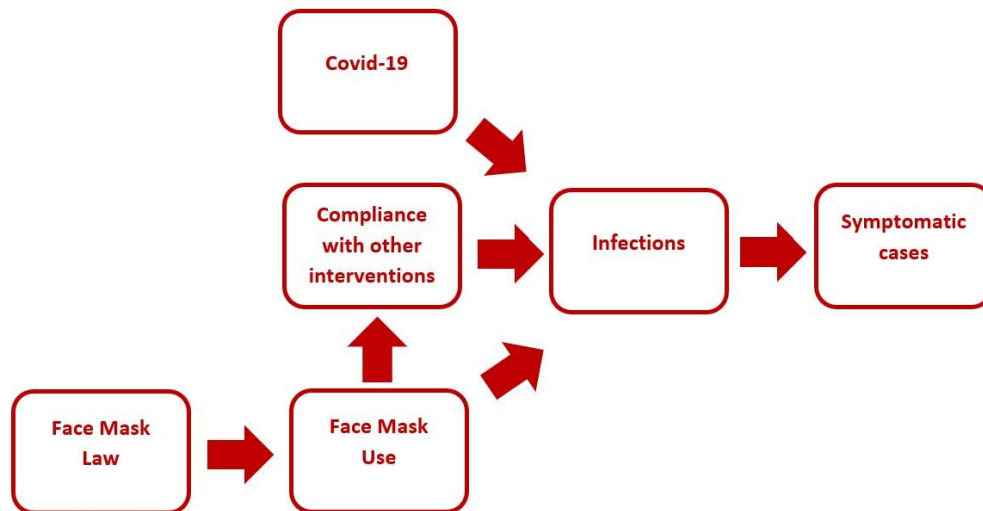
2. Identify specific mechanism hypotheses:

A plausible mechanism hypothesis is that a legal requirement to wear a face mask in public increases the use of face masks which in turn reduces the prevalence of Covid-19 which reduces the prevalence of symptomatic covid infections and thereby the number of hospitalizations and deaths.

A plausible hypothesised counteracting mechanism is that a legal requirement to wear a face mask in public will decrease compliance with other public health interventions, such as social distancing. This in turn would result in an increase in the number of symptomatic infections compared to the number that would have occurred if the legal requirement to wear a face mask had not been introduced.

² Jefferson et al. (2023)

³ Greenhalgh et al., (2022); Howard et al., (2020).



3. Specify the correlation claim:

The correlation claim is that a legal requirement to wear face masks in public is negatively correlated with symptomatic Covid-19 infections, conditional on potential confounders. Potential confounders include Covid-19 test rates, other public health measures such as social distancing, underlying health conditions, individual mobility, population density, socio-economic factors, age, gender, temperature and humidity.

4. Identify and assess association studies:

An event study conducted in the U.S. that covered all 50 states found mask mandates are associated with a 14% decrease in Covid-19 cases, conditional on several potential confounders including Covid-19 test rates, other public health interventions, individual mobility and weather.⁴

An observational study of matched cohorts from 412 U.S. counties found mask mandates are associated with decreases in Covid-19 cases ranging from 11% to 25 % during a six-week post intervention period, conditional on potential confounders.⁵

Analysis of data covering 188 nations found that mask mandates are consistently associated with lower Covid-19 infection rates in the short term and early adoption of mask mandates is consistently associated with lower infection rates in the longer term, conditional on potential confounders.⁶

5. Assess the need for a mechanistic evaluation:

Taken together, the quantitative studies detect a robust correlation across contexts. It is therefore plausible to conclude that the correlation claim is established and that confidence in the general mechanistic claim is increased.

Although each of the quantitative studies controls for several confounders, some residual confounding may persist. The causal claim is therefore only deemed to be provisionally established. A mechanistic evaluation is necessary because it could increase confidence in the causal claim.

⁴ Adjodah et al., (2021).

⁵ Huang et al., (2022).

⁶ An et al., (2021).

6. Identify and assess mechanistic studies:

The mechanism hypothesis is that a legal requirement to wear a face mask will increase face mask use and thereby decrease the prevalence of Covid-19 symptomatic infections, hospitalizations and deaths.

The first part of the mechanism hypothesis is that a legal requirement to wear face masks will increase face mask use. Analysis of survey data found that mask wearing increased up to 23.4% after the implementation of State mandates in Hawaii, Iowa, North Dakota and New Hampshire.⁷ Betsch et al. conducted cross-sectional surveys with 6973 German participants. They found that the implementation of a public mask mandate steeply increased mask wearing. They also found that sufficient compliance would unlikely be achieved with a voluntary mask policy.⁸ MacIntyre et al. conducted cross sectional surveys in Sydney and Melbourne (Australia), London (UK) and Phoenix and New York (USA). They found mask mandates are a predictor of mask wearing.⁹

The second part of the mechanism hypothesis is that face mask use will reduce the spread of Covid and thereby reduce symptomatic infections, hospitalizations and deaths. A recently updated Cochrane review found face masks make little or no difference to the spread of Covid-19.¹⁰ In contrast, multivariate analysis of 196 countries found that duration of mask wearing is negatively associated with Covid-19 mortality and that cultural norms and policies supporting public face mask wearing are associated with lower per-capita coronavirus mortality.¹¹ Model simulations found widespread use of face masks effectively reduces community transmission and decreases hospitalizations and deaths.¹² Experimental studies provide support for the efficacy of face masks in reducing transmission by supporting aerosol transmission as a key mode of transmission.¹³

The hypothesised counter-acting mechanism is that a legal requirement to wear face masks increases Covid-19 symptomatic cases by decreasing compliance with other public health interventions such as social distancing. Surveys found mask wearing is positively correlated with complying with other public health interventions, such as social distancing.¹⁴

By combining a range of quantitative and qualitative studies, evidence for each section of the mechanism hypothesis can be provided.

The studies relating to the first part of the mechanism hypothesis all support the effectiveness of a legal requirement in increasing face mask wearing. The studies provide evidence across contexts and control for a number of confounders. However, the studies rely on self-reported survey data and therefore there is a risk of recall bias and response bias in the results. Furthermore, although the studies cover a range of geographical regions, they are not fully representative of the global population.

Turning to the second part of the mechanism hypothesis, the Cochrane review included only RCTs and, as the authors acknowledge, the observed lack of effectiveness could be due to study design flaws. Furthermore, most of the studies included in the review evaluated masks in terms of how well

⁷ Adjodah et al., (2021).

⁸ Betsch et al. (2020).

⁹ MacIntyre et al., (2021).

¹⁰ Jefferson et al., (2023).

¹¹ Leffler et al., (2020).

¹² Eikenberry et al., (2020).

¹³ Van Doremalen et al., (2020); Leung et al., (2020).

¹⁴ Betsch et al., (2020); MacIntyre et al., (2021).

they protect the user rather than how well they reduce community transmission and therefore focus on a different outcome of interest to the outcome of interest here. Given these limitations, the results of the systematic review should be taken as inconclusive rather than negative in relation to the effectiveness of face masks in reducing the spread of Covid-19.

Taking account of a broader range of evidence enables a more accurate assessment of the effectiveness of face masks. The other studies considered here provide support for the effectiveness of face masks in reducing community transmission and thereby reducing symptomatic cases, hospitalizations and deaths. Taking account of a broader range of evidence, Greenhalgh et al. conclude that airborne aerosol transmission 'is likely to be dominant'.¹⁵ A narrative literature review concluded that face masks are effective at reducing community transmission when compliance is high.¹⁶

The survey results undermine the counteracting mechanism hypothesis. The surveys provide evidence across contexts and control for confounders but there remains a risk of bias in the survey results.

7. Assess the specific mechanism hypothesis:

The studies considered here provide support for the first part of the mechanism hypothesis. Some risk of bias remains, however, and the results are not representative of the global population. Therefore, the first part of the mechanism hypothesis is provisionally established.

The evidence considered here provides strong support for the second part of the mechanism hypothesis. The limitations of the systematic review mean it is inconclusive and does not provide high-quality evidence against the second part of the mechanism hypothesis. Taken together, the rest of the evidence considered here provides strong support for the effectiveness of face masks, particularly when widespread adoption of face masks is achieved. Therefore, the second part of the mechanism hypothesis is established.

The survey results provide evidence against the counteracting mechanism. Furthermore, given the extent of the net benefit of a legal requirement to wear face masks found across multiple studies, it is evident that the benefit is not negated, even if slightly reduced. The counteracting mechanism hypothesis is, therefore, arguably false.

Although the second part of the mechanism hypothesis is established, the first part is only provisionally established. The status of the mechanism hypothesis is that of its weakest part. Therefore, on the basis of the quantitative and qualitative evidence considered here, the mechanism hypothesis is provisionally established overall.

8. Assess the correlation claim:

The correlation claim is now established. The quantitative and qualitative evidence considered provides strong evidence of both correlation and mechanisms. The evidence of mechanisms increases confidence in the correlation claim to such an extent that it is unlikely that further evidence would overturn it.

9. Assess the general mechanism claim:

The evidence of mechanisms provisionally establishes the specific mechanism hypothesis. However, the strength of the quantitative evidence of correlation further increases confidence in the general

¹⁵ Greenhalgh et al., (2021).

¹⁶ Howard et al., (2021).

mechanistic claim. Given the mutually supporting evidence of correlation and mechanism, the general mechanistic claim is deemed to be established.

I0. Infer the status of the causal claim:

On the basis of the quantitative and qualitative evidence considered here, both the correlation claim and the general mechanistic claim are established. Therefore, the causal claim is established.

5. Example: wealth inequality and poverty

In this section, we consider the claim that wealth inequality causes poverty in the UK, in order to provide an example in which the cause under investigation is not an intervention and in which there are multiple interacting mechanisms.

Correlation claim

There is a wide range of evidence in support of the claim that wealth inequality is positively correlated with poverty in the UK, with particularly strong examples including:

The 2017 LSE report commissioned by Oxfam, [Double Trouble: A review of the relationship between UK poverty and economic inequality](#), demonstrates that both income inequality and wealth inequality are correlated with poverty in the UK, and indeed have been since at least the 1960s.

The 2022 Resolution Foundation report [Arrears Fears: The distribution of UK household wealth and the impact on families](#) demonstrates the positive correlation between wealth and poverty, and suggests that wealth acts as a mechanism that prevents families from falling into poverty.

The 2025 Joseph Rowntree Foundation [UK Poverty 2025](#) report demonstrates a correlation between the worsening rate of poverty in the UK and the continued unequal distribution of wealth.

Taken together, this evidence establishes that wealth inequality and poverty are correlated in the UK.

Mechanism hypotheses

Wealth inequality gives a small population in UK society the ability to either directly participate in, or indirectly prolong, wealth extraction practices. These wealth extraction practices further increase wealth at the top of the economic spectrum and drive the larger lower end of the economic spectrum into poverty.

There are a wide range of practices and processes that constitute the hypothesised mechanism complex linking wealth inequality to poverty. These are portrayed in Figure 2. What is crucial here is that the initial presence of wealth inequality *directly facilitates* those with wealth to engage in the extractive practices that in turn drive rates of poverty. As such, wealth inequality cannot be ignored: it is a key mechanism driving the extractive practices that cause poverty to rise.

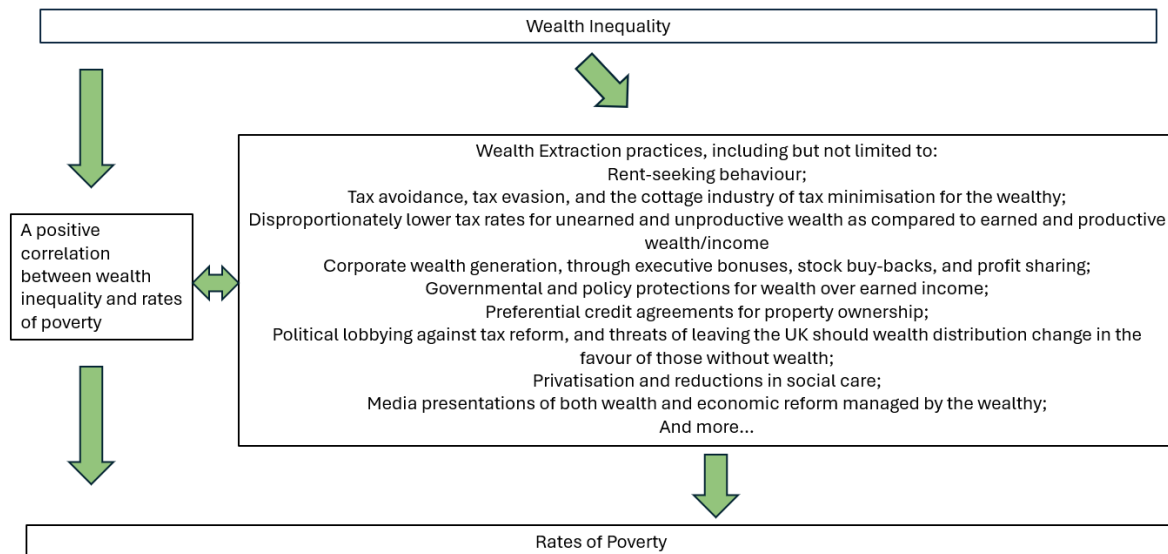


Figure 2. Mechanisms linking wealth inequality to poverty.

There are a wide range of specific mechanism hypotheses at play in this mechanism complex. The examples provided below demonstrate the strength of evidence available at the time of our review, rather than the importance of the mechanisms themselves.

Tax rates

As a direct result of the unequal distribution of wealth, a small wealthy subpopulation of the UK benefits from—and perpetuates—preferential tax rates for unearned and unproductive wealth.

This includes:

Lower taxes on unearned and unproductive wealth (such as capital gains tax and inheritance tax) compared to income tax (see the Wealth Tax Commission’s 2020 paper ‘[Sources of wealth and their implications for taxation](#)’; Advani & Summers’ 2020 paper ‘[Capital Gains and UK Inequality](#)’; and Harbury and Hitchen’s 1979 book ‘[Inheritance and Wealth Inequality in Britain](#)’);

The capacity for the wealthy to participate in political lobbying, particularly against tax reform and the introduction of a wealth tax in the UK (see the 2025 Joseph Rowntree Foundation report ‘[Talking about Wealth Inequality](#)’; CIPR’s 2025 report ‘[No Rules Britannia? The UK’s lobbying laws exposed on the global stage](#)’; and Solaiman’s 2023 paper ‘[Lobbying in the UK: Towards Robust Regulation](#)’);

The use of tax reduction/avoidance to protect and further grow wealth for those that already have it (see Equality Trust’s 2022 report ‘[Billionaire Britain](#)’; the [UK Government’s own 23/24 Tax Gap statistics](#); and Tax Justice UK’s claim that [the £47bn tax gap is itself an underestimation](#)).

A further implication is that the overall reduction in the tax budget as a direct result of these actions has further knock-on effects on the availability and quality of tax-funded social, economic, and health care available to those without wealth.

We found that there is sufficient high-quality evidence to conclude that this specific mechanism hypothesis is established.

Rent-seeking

Those at the top of the wealth distribution spectrum are able to engage in and propagate rentier capitalism ('rentierism') and rent-seeking behaviours, which in turn worsens the economic wellbeing of those without wealth and further increases the risk of falling into poverty and housing-instability for those without wealth.

This includes:

The wealthy engaging in rent-seeking behaviours (see Christophers' 2019 paper '[The rentierization of the United Kingdom economy](#)'; Toms and Lin's 2020 paper '[Rent-Seeking, Financial Scandals and Inequality: Evidence from the UK, 1908-2010](#)'; Byrne's 2018 paper '[Generation rent and the financialization of housing](#)');

Those behaviours unfairly affecting those without wealth (see Angelopoulos et al's 2021 paper '[The distributional consequences of rent-seeking](#)'; the 2023 IFS report '[Housing costs and income inequality in the UK](#)'; and McKee et al's 2019 paper '[Generation rent and the emotions of private renting: self-worth, status and insecurity amongst low-income renters](#)').

We found that there is sufficient high-quality evidence to conclude that this specific mechanism hypothesis is provisionally established.

Regional differences in UK wealth distribution

Regional differences in wealth further drive rates of poverty, both locally and across the country. The disproportionate distribution of wealth towards the south of the UK drives rates of poverty in the North of England and the devolved nations, and further reduces the overall wealth in those areas. It also increases the rates of poverty in the areas that have high wealth (particularly London), because the wealth is not shared equally in the areas of high concentration.

This includes:

Differences in wealth between London and the South East on the one hand, and the North East and Wales on the other (see the 2025 Joseph Rowntree Foundation '[UK Poverty](#)' report; the ONS' 2022 report '[Distribution of individual total wealth by characteristic in Great Britain: April 2018 to March 2020](#)'; and the Equality Trust's 2023 report '[The Scale of Economic Inequality in the UK](#)')

Local wealth inequality in London drives high poverty rates within London itself (see Trust for London's 2023/24 '[Poverty by Region report](#)'; ONS' 2021 report '[What are the regional differences in income and productivity?](#)'; and London Datastore's 2022 report on '[Economic Fairness: Population in Poverty](#)').

There is a further dimension of ethnicity to this geographical inequality and poverty, whereby those from ethnic minority backgrounds in the UK have disproportionately less wealth than white Britons (see Karagiannaki's 2023 paper '[The scale and drivers of ethnic wealth gaps across the wealth distribution in the UK: evidence from Understanding Society](#)'; Roberts' 2023 paper '[London's poverty rate is shockingly and stubbornly high](#)'; and the 2022 Joseph Rowntree Foundation report '[Ethnicity and the heightened risk of very deep poverty](#)').

We found that there is sufficient high-quality evidence to conclude that this specific mechanism claim is provisionally established.

Monopoly Power

Monopoly power increases poverty by driving up prices, increasing the ability of the wealthy to engage in political lobbying, and reducing competition in the economic market. Moreover, wealth inequality is a driver of monopoly power, as those with wealth can engage in practices to both participate in and protect monopoly power.

This includes:

The macroeconomic effects of monopolies on poverty and economic inequality, whereby monopolistic companies are disincentivised from paying higher wages, producing higher quality goods, and lowering prices (see the 2023 Global Justice Now report [‘Monopoly capitalism: What is it and how do we fight it?’](#); and the 2022 IFS paper [‘What drives wage stagnation: monopsony or monopoly?’](#))

The political and lobbying influence that monopoly affords those with wealth to push deregulation and prevent tax reform, and to protect the current economic conditions that drive poverty (see Zephyr Teachout’s 2018 paper [‘The Problem of Monopolies & Corporate Public Corruption’](#); and Zia, Farooq, and Al Ajmi’s 2021 paper [‘Finding the Impact of Market Visibility and Monopoly on Wealth Distribution and Poverty Using Computational Economics’](#))

The historical and colonial history of monopoly power creation and protection, whereby those who have colonially and/or historically generated familial wealth are able to protect and grow their own wealth, while limiting the wages and wealth of their workers (see Oxfam’s 2025 report [‘Takers, not Makers: The unjust poverty and unearned wealth of colonialism’](#))

We found that there is sufficient high-quality evidence to conclude that this specific mechanism hypothesis is provisionally established.

These four specific mechanism hypotheses provide strong evidence for the general mechanistic claim, that the correlation between wealth inequality and poverty is attributable to a network of extractive wealth practices by which the wealth grow and protect their own wealth, and which lead to increased poverty rates as a direct result.

On the basis of our review, we classified the causal relationship between wealth inequality and poverty as provisionally established.

Given that the causal link from wealth inequality to poverty is provisionally established, there is a need for policymakers to explore ways to reduce wealth inequality as a means to reduce poverty.). We consider universal basic income in more detail in section 8 below. [Wealth Gap Risk Register 2025](#)). We consider universal basic income in more detail in section 8 below.

6. Example: interventions to tackle fake news

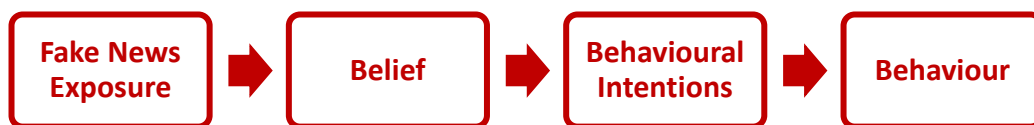
In the following four sections we provide examples of how Evidential Pluralism offers a helpful framework for identifying evidence gaps and how best to fill them.

The Problem

It is often assumed that online fake news has a detrimental impact on behaviour, but is this really the case? According to EP, to establish that online fake news affects behaviour, one needs to establish the existence of both a correlation and a mechanism.¹⁷ This requires combining quantitative and qualitative evidence.

Quantitative studies provide evidence of correlation. For example, there is evidence of a correlation between the amount of misinformation shared and a decrease in daily Covid-19 vaccination rates, conditional on potential confounders.¹⁸

A plausible mechanism hypothesis connecting fake news and problematic behaviour is:



Qualitative and quantitative studies provide evidence of this mechanism.¹⁹ For example, qualitative surveys identify volume of information, repeated exposure, using family and friends as sources of information and emotive content as drivers of belief in fake news.²⁰ An Experimental study found exposure to misinformation is associated with small but significant changes in behavioural intentions and that behavioural intentions are associated with belief.²¹

Thus, by combining quantitative evidence of correlation and quantitative and qualitative evidence of mechanisms, the detrimental effects of online fake news can be established.

Interventions

The above mechanism hypothesis and associated evidence can be used to identify possible points of intervention. For example, accuracy prompts aim to reduce the impact of fake news by improving truth discernment which (i) reduces belief in fake news²² and (ii) reduces sharing of, and in turn exposure to, fake news.²³

¹⁷ 'Fake news' is here being used to cover all kinds of false and misleading information, including misinformation, disinformation and mal-information.

¹⁸ Pierri et al. (2022).

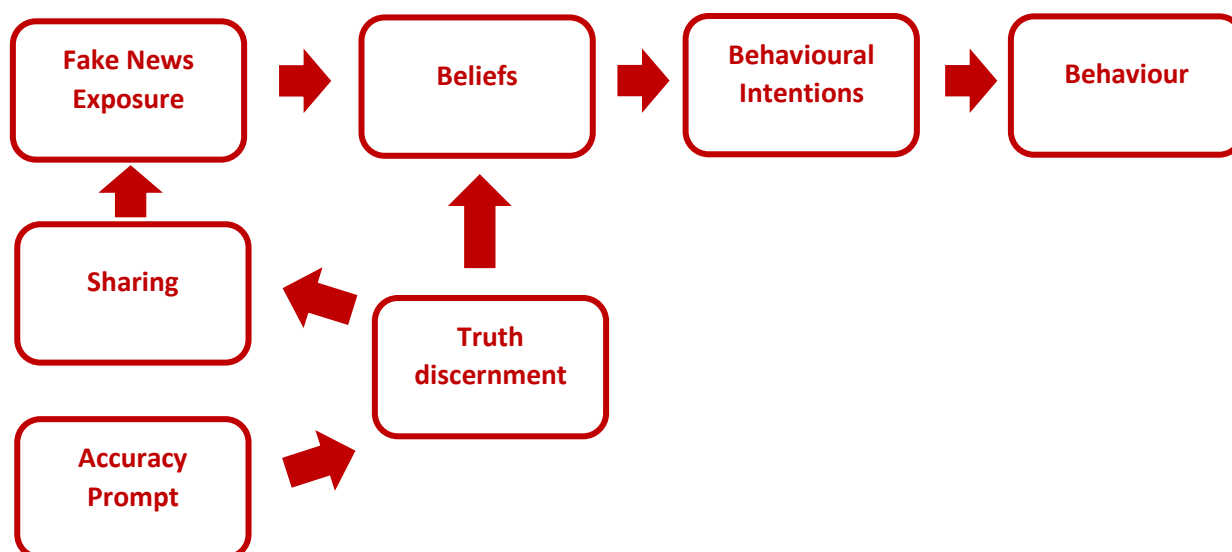
¹⁹ See, for example, Allington et al. (2021); Greene and Murphy (2021); Lockyer et al. (2021); Pennycook et al (2018); Roozenbeek et al. (2020)

²⁰ Lockyer et al. (2021).

²¹ Greene and Murphy (2021).

²² There is evidence that belief in fake news is driven by failing to engage in reflective reasoning and instead relying on automatic, intuitive thinking. See, for example, Bago et al. (2020); Pennycook and Rand (2019); Pennycook and Rand (2021).

²³ There is evidence that repeated exposure increases belief in fake news. See, for example, Pennycook et al. (2018).



Evidence of mechanisms is also crucial to evaluating interventions. Quantitative and qualitative studies provide evidence of mechanisms to support the effectiveness of accuracy prompt interventions. For example, survey studies show that accuracy prompts increase truth discernment and reduce the amount of false content people intend to share online.²⁴ A large field experiment conducted on Twitter shows that accuracy prompts reduce the amount of false content shared online.²⁵ Computational modelling provides evidence that accuracy prompts increase truth discernment and improve quality of content shared by focusing attention on accuracy rather than increasing the amount of deliberation.²⁶

Combining quantitative and qualitative evidence provides strong evidence of mechanisms to support the effectiveness of accuracy prompt interventions. This evidence can be understood to provide indirect evidence of correlation via channel m_3 in Fig. 1 above. Quantitative evidence of a correlation between accuracy prompts and behaviour, such as improved Covid-19 vaccine uptake, would further support the effectiveness of accuracy prompts.

7. Example: interventions to reduce consumption of alcohol

The problem

Does alcohol consumption cause cancer? EP captures the evidence appraisal procedures employed by the International Agency for Research on Cancer (IARC) to answer questions such as this. The IARC *Monographs* programme evaluates the strength of the evidence in order to determine whether a particular preventable exposure can be classified as a possible, probable, or established cause of

²⁴ Arechar et al. (2022); Brashier et al. (2020); Epstein et al. (2021); Fazio (2020); Pennycook et al. (2020); Pennycook et al. (2021); Pennycook and Rand (2022).

²⁵ Pennycook et al. (2021); Pennycook and Rand (2022).

²⁶ Lin, Pennycook and Rand (2023).

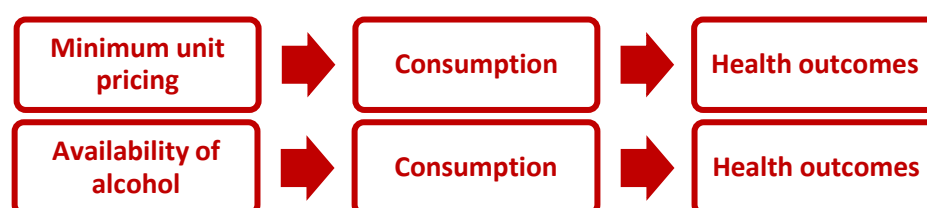
cancer in humans. IARC determined that consumption of alcoholic beverages does indeed cause cancer.²⁷

A *Monographs* working group evaluates a range of evidence in order to determine whether an exposure is a cause of cancer.²⁸ One subgroup evaluated evidence of cancer in human epidemiological studies relating to alcohol consumption. These quantitative studies provided strong evidence of the existence of a correlation between alcohol consumption and cancer in humans. But such studies alone are often insufficient to establish that the exposure is a cause of cancer, due to the risk of confounding and bias. Another subgroup therefore evaluates the evidence from mechanistic studies in order to determine whether there is a mechanism linking exposure to cancer. These studies can be quantitative or qualitative. There was substantial mechanistic evidence that alcohol causes cancer of the oesophagus, for example.

Interventions

Establishing that alcohol consumption is a cause of cancer and other diseases has led to a number of interventions aimed at reducing alcohol intake. One such intervention is the introduction of minimum unit pricing, that is, a legal minimum price per unit of alcohol. EP can help to assess the effectiveness of such an intervention on different populations.

Scotland introduced minimum unit pricing in 2018, but there is currently no minimum unit pricing in England. Quantitative studies comparing England and Scotland have shown a population-level correlation between the introduction of minimum unit pricing and a reduction in deaths and hospitalisations.²⁹ One possible explanation of this correlation appeals to the familiar economic mechanism by which an increase in price leads to a fall in demand and therefore consumption of alcohol. However, there are also alternative mechanism hypotheses. For example, the Scottish population in which minimum unit pricing was introduced may have had an unrelated, simultaneous drop in the availability of alcohol relative to the English control population; it may be a drop in the availability of alcohol, rather than the minimum unit pricing, that is causing most of the reduction in deaths and hospitalisations.³⁰



Qualitative and quantitative studies can help to decide between the two competing mechanism hypotheses. For example, quantitative studies help to disconfirm the availability hypothesis by providing evidence of a similar level of availability of alcohol in Scotland and England.³¹ And qualitative or mixed methods studies, for example, studies including structured interviews, help to determine whether it is the affordability or the availability of alcohol that is causing the reduced

²⁷ IARC (2010).

²⁸ IARC (2019).

²⁹ See, for example, Wyper et al. (2023).

³⁰ See, for example, PHS (2023) pp. 60-77.

³¹ PHS (2023), pp. 72-74.

consumption.³² In this way, integrating quantitative and qualitative evidence can help to establish the effectiveness of a proposed intervention.

8. Example: universal basic income

EP can help to identify gaps in the evidence base, as in the example of Universal Basic Income (UBI). Where the quantitative evidence gives some inconclusive and low-quality support to the claim that UBI is correlated with health and economic benefits, EP suggests that there is a need for additional mechanistic evidence, rather than only additional costly quantitative pilot studies.

UBI is purported to have positive economic effects, by reducing economic inequality and precarity³³, and positive public health effects, by reducing the rate of depressive disorders and preventable long-term conditions, and the subsequent cost on the NHS.³⁴

A number of small-scale trials have been piloted to assess the effectiveness of UBI in the UK: a Welsh pilot study launched in 2022 tested the effectiveness of a monthly £1600 stipend for the wellbeing of 635 people leaving care over 2 years;³⁵ two trials were proposed in 2023 for areas in the North and South of England, in which 30 people would also be given £1600 a month for 2 years;³⁶ and UBI is a point of political discussion in Scotland³⁷ and Ireland.³⁸

However, these micro-trials have limitations: due to their high cost, trial populations are very small, often limited to specific geographic areas and demographics; and while they contain control groups, trials can't be properly randomised or double-blinded, as participants know if they're receiving the stipend or not. Thus, they are imperfect trials and produce low-quality, inconclusive evidence.

There is a clear gap in the evidence base, that can be addressed in two ways: by conducting larger, higher quality experimental trials, and/or by seeking mechanistic evidence³⁹.

Larger and more diverse trials are costly and hard to justify in the absence of (provisional) evidence of effectiveness, and even the cost of the small pilots themselves are a barrier for policymakers: the 2022 trial in Wales is not being continued beyond its 2025 end date due to the cost, despite overall positive responses from participants⁴⁰; and the English pilots proposed in 2023 have still yet to launch, as Sadiq Khan argued that "[governmental financial] support is there... someone else would need to fund a pilot."⁴¹ Despite the economic benefits of UBI being modelled by the NHS⁴² and their affordability being rationalised within reasonable budgets⁴³, operating these trials, particularly at larger scales, is too expensive, unjustified, and risk for policymakers to readily adopt.

³² See, for example, Holmes et al. (2022).

³³ See, for example, Lowrey, 2018; Reed et al, 2023.

³⁴ See, for example, Gibson et al, 2020; Johnson et al, 2023.

³⁵ See Drakeford, 2022.

³⁶ See Ali Hussen, 2023.

³⁷ The Scottish Greens party made a 2024 commitment to pursuing UBI in Scotland: <https://greens.scot/news/scottish-greens-will-move-to-universal-basic-income>

³⁸ See Redmond et al, 2022.

³⁹ See Hoynes and Rothstein (2019) for further information on the limitations of these micro-trials.

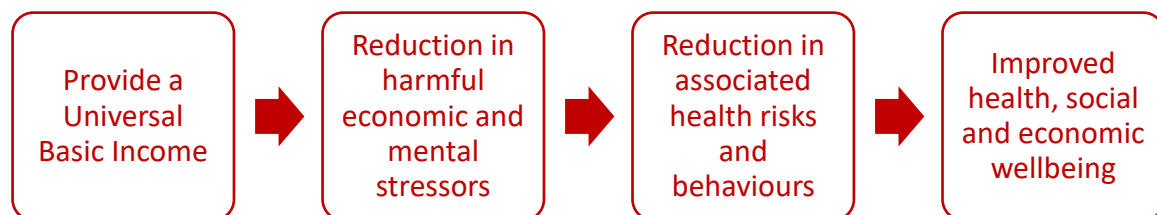
⁴⁰ Gwilym and Beck, 2023.

⁴¹ See Khan, 2023

⁴² Johnson et al, 2023.

⁴³ Reed et al, 2023.

Rather than investing heavily in scaling up trials, a more cost-effective approach would be to integrate mechanistic evidence with the existing quantitative evidence.⁴⁴⁴⁵ There is currently limited mechanistic evidence for both the positive and negative effects of UBI on participation in the labour market, limited mechanistic modelling of how UBI could be funded and its impact on national budgets, and a limited mechanistic understanding of how UBI reduces low-income related stress and health issues.⁴⁶ There is a clear gap in the evidence base in this regard, and more research is needed to fully explore the hypothesised mechanism underpinning UBI as a health and economic intervention, which can be summarised as follows:



By investigating this proposed mechanism hypothesis through a range of quantitative and qualitative mechanistic studies that provide an understanding of how UBI reduces low-income stress and related health issues, policy makers will be in a better position to evaluate whether and why UBI works or not, and would have higher quality evidence to justify larger scale UBI trials.

Given the public interest in the proposed health⁴⁷ and economic benefits⁴⁸ of UBI, there is a clear need for more confidence in the effectiveness of UBI as a policy intervention, as well as greater accountability and understanding. While larger and more diverse trials are desirable, obtaining mechanistic evidence is more cost-effective and less risky in this case, and EP provides a practical method for integrating quantitative and qualitative mechanistic evidence in the evaluation of UBI.

9. Example: awarding gaps in higher education

In the UK in 2020-2021, the gap between the percentage of white students and the percentage of black students receiving a first or 2:1 degree was 18.5%. The gap between white students and Asian students was 6.1% and the gap between white students and students from all other ethnicities was 9%.⁴⁹

⁴⁴ Johnson et al, 2021.

⁴⁵ This has been noted by Hoynes and Rothstein (2019), who argue that: “Unfortunately, the planned and ongoing pilots are not well suited to answer these questions [of effectiveness]. Experimentation aimed at identifying parameters and mechanisms... would be more useful than evaluations of small UBI pilots” (p. 24, my own bracketing).

⁴⁶ See, for example, Fitzpatrick, 2022; Jaimovich et al, 2022.

⁴⁷ See, for example, Ruckert et al, 2017; Johnson et al, 2023.

⁴⁸ See, for example, Srnicek & Williams, 2015; Harrop & Tait, 2017; Lowrey, 2018.

⁴⁹ Advance HE (2022).

Addressing these inequalities requires understanding their causes. According to EP, this requires evidence of correlation and evidence of mechanisms.

Large quantitative studies provide evidence of a correlation between ethnicity and degree outcomes, conditional on potential confounders including prior qualifications, social class, and other demographics.⁵⁰

Qualitative research has identified several underlying factors that might explain the correlation between ethnicity and degree outcomes, including previous educational experience, curriculum design and content, teaching, learning and assessment approaches, educational environment, and direct and indirect racism.⁵¹

However, as Sabri has argued, these findings are not sufficiently discriminating. To enable effective change, more nuanced analysis is needed of the strength of different causes and their interaction in different contexts.⁵² This requires explicitly formulating and assessing mechanism hypotheses.

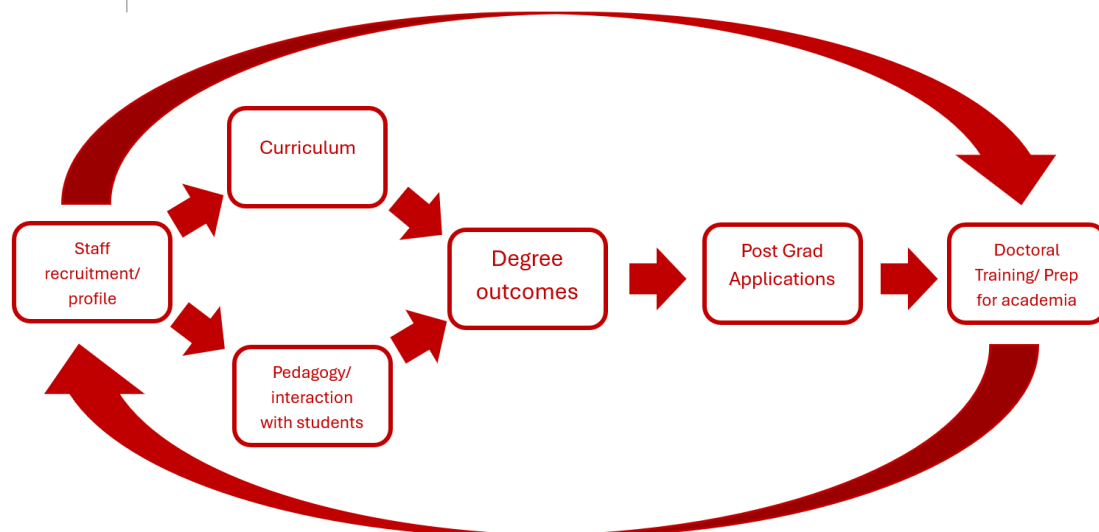
As Sabri notes, there is an assumption in public discourse that greater representation of Black and minority ethnic staff among academics will reduce the inequality in degree outcomes. We should not assume, however, that greater representation among staff would have a simple, positive causal effect on degree outcomes. A plausible mechanism hypothesis that Sabri identifies as embedded in the literature is that under-representation of Black and ethnic minority staff limits diversity in expertise which limits diversity in curricula design.⁵³ This, in turn, leads to a gap between curricula content and the interests of an increasingly diverse student population which affects student degree outcomes. The inequality in degree outcomes perpetuates the under-representation of Black and ethnic minority staff. The under-representation of Black and ethnic minority students with 'good degrees' results in a disproportionately white population of students from which a small number are recruited to undertake PhDs and train for an academic career. An additional mechanism hypothesis discussed by Sabri is that high staff-student ratios facilitate greater engagement with students and provision of formative feedback, which in turn enable students to fulfil their potential. These mechanisms and their interactions are captured in the following diagram.

⁵⁰ See, for example, Connor et al. (2004); Richardson (2015); HEFCE (2015).

⁵¹ Brunce et al. (2021); Singh (2011).

⁵² Sabri (2023). Sabri provides a full exploration of the benefits of applying Evidential Pluralism to understanding and addressing inequalities in degree outcomes.

⁵³ Sabri (2023).



Both quantitative and qualitative studies can provide evidence of the extent of the contribution of these hypothesized mechanisms. For example, quantitative studies could assess the relationship between staff diversity and student degree outcomes. Qualitative studies could assess other stages of the mechanism. For example, analysis of survey results suggest that non-white staff are more likely to include diversity related content in their curricula.⁵⁴

Systematically assessing evidence of correlation from quantitative studies and evidence of mechanisms from quantitative and qualitative studies will enable greater understanding of the causes of inequality in degree outcomes. This, in turn, will enable the identification of potentially effective interventions that can in turn be evaluated by providing both evidence of correlation and evidence of mechanisms.

10. Further resources

Introductory material on Evidential Pluralism can be found at: blogs.kent.ac.uk/evidential-pluralism/

For more details on the evaluation methodology and face-masks example, see Trofimov, A & Williamson, J. (2025), Applying Evidential Pluralism to evidence-based law: EBL+, *Jurisprudence* 16(4):647-690. Open access at <https://dx.doi.org/10.1007/s10670-022-00611-0>

For a general account of the application of Evidential Pluralism to the social sciences, see: Shan, Y. and Williamson, J. (2023). *Evidential Pluralism in the Social Sciences*. Routledge, Abingdon. Open-access at <https://www.taylorfrancis.com/books/oa-mono/10.4324/9781003143000/>

11. Glossary

Evidential Pluralism. A theory of how to evaluate interventions and other causal relationships. According to this approach, it is important to assess mechanistic studies alongside experimental and

⁵⁴ See, for example, Mayhew and Grunwald (2006).

observational studies of the covariation between the putative cause and effect. Evidential Pluralism provides general guidance as to exactly how to do this.

Impact evaluation. An evaluation of whether a particular intervention is effective in achieving certain goals. An impact evaluation may also try to estimate exactly how effective the intervention is.

Mechanistic study. This kind of study provides information about the ways in which a cause produces its effects. A mechanistic study of an intervention may shed light on the process by which the intervention works, or on other processes that can interact to change the results of the intervention. Mechanistic studies shed light on key features of mechanisms, such as mediating variables, entities or activities involved in the mechanism, or the structure or spatiotemporal organisation of the mechanism.

Association study. This kind of study tests whether a putative cause and effect are probabilistically dependent, conditional on a set of potential confounders. Typically, such a study will also attempt to estimate the extent of this dependence. Association studies include studies such as RCTs as well as observational studies such as cohort studies. In the literature on Evidential Pluralism, these studies are often classified as association studies, because they are principally measuring the association between putative cause and effect, conditional on potential confounders.

Process evaluation. An evaluation of whether a particular intervention has been implemented as intended.

RCT. Randomised Controlled Trial. A trial that randomly allocates participants to the intervention. Those who are not given the intervention are usually given an alternative intervention, which will ideally be indistinguishable from the intervention but include none of the features of the intervention that are likely to cause the outcome of interest.

Realist evaluation. Realist evaluation looks at mechanisms of an intervention to ask for whom the intervention works and in which contexts it works. While Evidential Pluralism, in common with standard impact evaluation methods, asks whether there is an average effect across a whole population, realist evaluation can be thought of as focussing on variability in effect. For this reason, realist evaluation explores the fine-grained details of intervention mechanisms, while Evidential Pluralism looks for key features of a mechanism, and thus appeals to more coarse-grained mechanism hypotheses.

Theory-based evaluation. This approach to evaluation focuses on the importance of theories to evaluation. Interventions are often represented using ‘theories of change’ or ‘logic models’, in order to structure an evaluation. These representations are viewed as kinds of specific mechanism hypothesis by Evidential Pluralism. Evidential Pluralism has a different emphasis: it views theories as having no evidential value on their own—only when confirmed by evidence can a theory be relevant to an evaluation. Thus Evidential Pluralism focuses on evidence, rather than theory.

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